6.5.4 Hydrological Cycle

Continuous movement of water among the three spheres is known as **Hydrological Cycle**. Hydrological cycle involves evaporation, condensation, precipitation, advection, interception, evapo-transpiration, infiltration, percolation and runoff to the ocean (Figure 6.23).

Evaporation is the process by which water in liquid state changes into vapour state using heat energy from Sun. Evaporation is maximum when the temperature is high, on the large expanse of water and when dry winds blow over water surface.

Condensation is the process by which water vapour cools to form water droplet by loosing temperature. The condensation occurs when dew point is reached in the atmosphere.

Precipitation is the process by which all forms of water particles fall from the atmosphere and reach the ground.



The rain drop that falls may get evaporated before it reaches the ground in an extremely arid region.

6.6 Precipitation

Precipitation is the product of condensation of atmospheric water vapour that falls under gravity and reaches the surface of the earth. In order to fall as rain drop or snow, the tiny drop lets in a cloud must grow larger. The droplets accumulate over the nuclei and combine to grow large enough to fall and reach the surface of the earth due to gravity.

If the drop is smaller it falls slowly so that it evaporates before it reaches the ground. Ice crystals in cloud also cause precipitation. Each ice crystal grows by cooling so that they become large in size and fall to the ground. They melt on the way due to friction with the atmosphere and fall as rain.

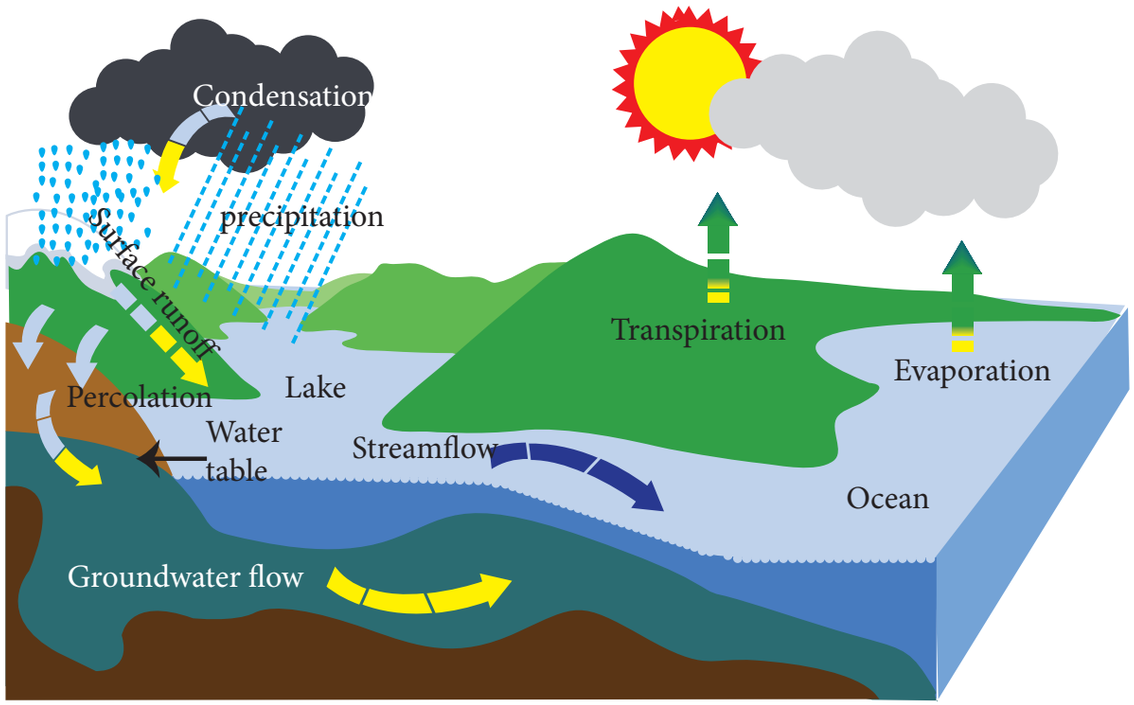


Figure 6.23 Hydrological cycle

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